

MERIDIAN UNIVERSITY · DEPARTMENT OF PHYSICS

Quantum Mechanics I

Problem Set 4: Operator Algebra, Spin Dynamics & Perturbation Theory

Course: PHYS 401

Term: Spring 2026

Instructor: Prof. Elena Vance

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Due Date: March 27, 2026

Student Name: _____

Student ID: _____

Total Points: 100 Points

Format: Show all intermediate operator derivations.

Honor Code: Standard academic integrity rules apply.

Problem 1: Harmonic Oscillator Matrix Elements & Uncertainty [30 Points]

Consider a one-dimensional quantum harmonic oscillator of mass m and natural frequency ω . Recall the ladder (annihilation and creation) operators defined by:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right), \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right)$$

which obey the canonical commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$. The energy eigenstates are denoted by $|n\rangle$ with $n = 0, 1, 2, \dots$, where $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$.

- Express the position operator $\hat{x}$ and momentum operator $\hat{p}$ in terms of $\hat{a}$ and $\hat{a}^\dagger$. Use these to evaluate the matrix elements $\langle n|\hat{x}^2|m\rangle$ and $\langle n|\hat{p}^2|m\rangle$ for arbitrary non-negative integers n and m . [12 Points]
- Calculate the expectation values $\langle \hat{x} \rangle$, $\langle \hat{x}^2 \rangle$, $\langle \hat{p} \rangle$, and $\langle \hat{p}^2 \rangle$ for a stationary state $|n\rangle$. Show that the product of uncertainties $\Delta x \Delta p$ satisfies:

$$\Delta x \Delta p = \left(n + \frac{1}{2} \right) \hbar$$

Verify that the ground state $|0\rangle$ saturates the Heisenberg Uncertainty Principle bound. [10 Points]

- A quantum particle at $t = 0$ is prepared in the coherent superposition state:

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

Calculate the time-dependent expectation value of position $\langle \hat{x} \rangle(t)$ for $t > 0$, and determine its amplitude and oscillation frequency. [8 Points]

✓ Solution to Problem 1

Part (a): Operator Expressions and Matrix Elements

Inverting the definitions of $\hat{a}$ and $\hat{a}^\dagger$, we obtain:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger), \quad \hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}} (\hat{a} - \hat{a}^\dagger)$$